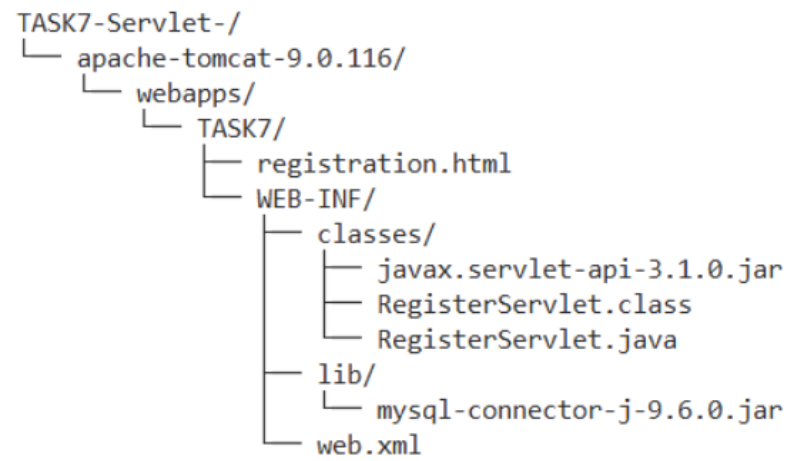


```

1  #registration.html
2  <!doctype html>
3  <html>
4      <head>
5          <title>User Registration</title>
6      </head>
7      <body>
8          <h2>User Registration</h2>
9
10         <form method="post" action="RegisterServlet">
11             <label>Username:</label><br />
12             <input type="text" name="name" required /><br /><br />
13
14             <label>Password:</label><br />
15             <input type="password" name="password" required /><br /><br />
16
17             <label>Email:</label><br />
18             <input type="email" name="email" required /><br /><br />
19
20             <label>Contact Number:</label><br />
21             <input type="text" name="contact" required /><br /><br />
22
23             <input type="submit" value="Register" />
24         </form>
25     </body>
26 </html>
27
28 #web.xml
29 <web-app>
30     <servlet>
31         <servlet-name>RegisterServlet</servlet-name>
32         <servlet-class>RegisterServlet</servlet-class>
33     </servlet>
34
35     <servlet-mapping>
36         <servlet-name>RegisterServlet</servlet-name>
37         <url-pattern>/RegisterServlet</url-pattern>
38     </servlet-mapping>
39
40 </web-app>
41 #RegisterServlet.java
42 import javax.servlet.*;
43 import javax.servlet.http.*;
44 import java.io.*;
45 import java.sql.*;
46
47 public class RegisterServlet extends HttpServlet {
48
49     protected void doPost(HttpServletRequest request, HttpServletResponse response)
50         throws ServletException, IOException {
51
52         response.setContentType("text/html");
53         PrintWriter pw = response.getWriter();
54
55         String name = request.getParameter("name");
56         String password = request.getParameter("password");
57         String email = request.getParameter("email");
58         String contact = request.getParameter("contact");
59
60         String url = "jdbc:mysql://localhost:3306/registration_DB_5R4";
61         String user = "root";
62         String pwd = "password";
63
64         try {
65             // ✅ Correct driver
66             Class.forName("com.mysql.cj.jdbc.Driver");
67
68             Connection con = DriverManager.getConnection(url, user, pwd);
69
70             // ✅ SQL matches form fields
71             String sql = "INSERT INTO students(name, password, email, contact) VALUES (?, ?, ?, ?)";
72             PreparedStatement ps = con.prepareStatement(sql);
73
74             // ✅ Use form data (NOT Scanner)
75             ps.setString(1, name);
76             ps.setString(2, password);
77             ps.setString(3, email);

```



```

78         ps.setString(4, contact);
79
80         int rows = ps.executeUpdate();
81
82         if (rows > 0) {
83             pw.println("<h2>Registration Successful!</h2>");
84         } else {
85             pw.println("<h2>Registration Failed</h2>");
86         }
87
88         con.close();
89
90     } catch (Exception e) {
91         pw.println("Error: " + e.getMessage());
92     }
93
94     pw.close();
95 }
96 }
97 #cookie.html
98 <!doctype html>
99 <html>
100     <head>
101         <title>Cookie Login</title>
102     </head>
103     <body>
104         <h2>Login using Cookies</h2>
105
106         <form action="login" method="post">
107             Username: <input type="text" name="username" required /><br /><br />
108             Password:
109             <input type="password" name="password" required /><br /><br />
110             <input type="submit" value="Login" />
111         </form>
112     </body>
113 </html>
114
115 #web.xml
116 <web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
117     version="3.1">
118
119     <servlet>
120         <servlet-name>cookieLogin</servlet-name>
121         <servlet-class>CookieLoginServlet</servlet-class>
122     </servlet>
123
124     <servlet-mapping>
125         <servlet-name>cookieLogin</servlet-name>
126         <url-pattern>/login</url-pattern>
127     </servlet-mapping>
128
129     <servlet>
130         <servlet-name>cookieRetrieve</servlet-name>
131         <servlet-class>CookieRetrieveServlet</servlet-class>
132     </servlet>
133
134     <servlet-mapping>
135         <servlet-name>cookieRetrieve</servlet-name>
136         <url-pattern>/retrieve</url-pattern>
137     </servlet-mapping>
138
139 </web-app>
140 #CookieLoginServlet.java
141 import java.io.*;
142 import javax.servlet.*;
143 import javax.servlet.http.*;
144
145 public class CookieLoginServlet extends HttpServlet {
146
147     protected void doPost(HttpServletRequest req, HttpServletResponse res)
148         throws IOException {
149
150         res.setContentType("text/html;charset=UTF-8");
151
152         PrintWriter out = res.getWriter();
153
154         String username = req.getParameter("username");

```



```

155     String password = req.getParameter("password");
156
157     // Validation
158     if (username == null || username.isEmpty() ||
159         password == null || password.isEmpty()) {
160
161         out.println("<h3 style='color:red;'>Invalid input! Please try again.</h3>");
162         return;
163     }
164
165     // Create cookies (INCLUDING password as per requirement)
166     Cookie userCookie = new Cookie("uname", username);
167     Cookie passCookie = new Cookie("pwd", password);
168
169     // Expiry
170     userCookie.setMaxAge(60 * 60 * 24); // 1 day
171     passCookie.setMaxAge(2 * 60); // 2 minutes
172
173     // Basic security improvement (still allowed)
174     userCookie.setHttpOnly(true);
175     passCookie.setHttpOnly(true);
176
177     res.addCookie(userCookie);
178     res.addCookie(passCookie);
179
180     out.println("<h2>Cookies Created Successfully</h2>");
181     out.println("<a href='retrieve'>Retrieve Cookie Details</a>");
182 }
183 }

```

```

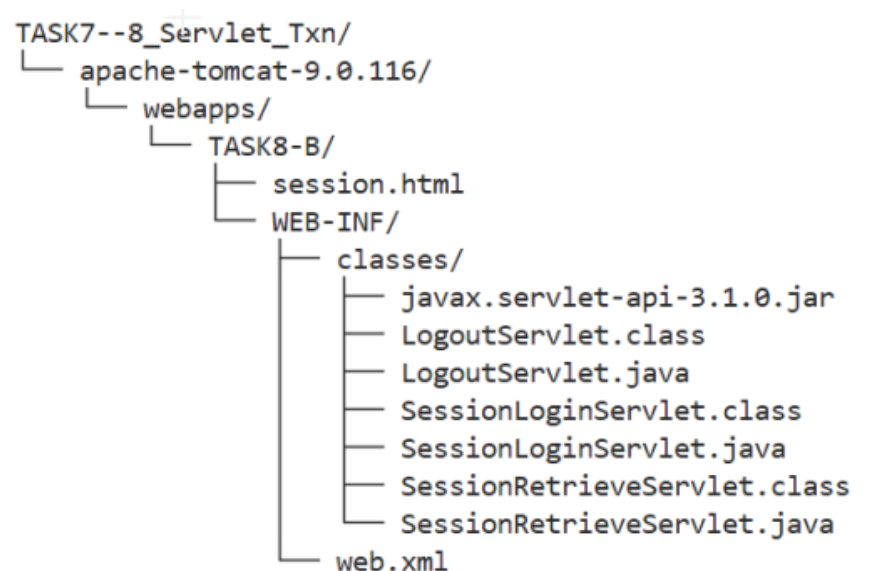
184 #CookieRetrieveServlet.java
185 import java.io.*;
186 import javax.servlet.*;
187 import javax.servlet.http.*;
188
189 public class CookieRetrieveServlet extends HttpServlet {
190
191     protected void doGet(HttpServletRequest req, HttpServletResponse res)
192         throws IOException {
193
194         res.setContentType("text/html;charset=UTF-8");
195
196         PrintWriter out = res.getWriter();
197
198         Cookie[] cookies = req.getCookies();
199
200         String username = null;
201         String password = null;
202
203         if (cookies != null) {
204             for (Cookie c : cookies) {
205
206                 if ("uname".equals(c.getName())) {
207                     username = c.getValue();
208                 }
209
210                 if ("pwd".equals(c.getName())) {
211                     password = c.getValue();
212                 }
213             }
214         }
215
216         if (username != null && password != null) {
217             out.println("<h2>Welcome " + username + "</h2>");
218             out.println("<h3>Password: " + password + "</h3>");
219         } else {
220             out.println("<h3>No cookies found. Please login first.</h3>");
221         }
222     }
223 }

```

```

224 #session.html
225 <!doctype html>
226 <html>
227     <head>
228         <title>Session Login</title>
229     </head>
230     <body>
231         <h2>Login using Session</h2>

```



```
232
233     <form action="login1" method="post">
234         Username: <input type="text" name="username" required /><br /><br />
235         Password:
236         <input type="password" name="password" required /><br /><br />
237         <input type="submit" value="Login" />
238     </form>
239 </body>
240 </html>
241
242 #web.xml
243 <web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee" version="3.1">
244
245     <servlet>
246         <servlet-name>sessionLogin</servlet-name>
247         <servlet-class>SessionLoginServlet</servlet-class>
248     </servlet>
249
250     <servlet-mapping>
251         <servlet-name>sessionLogin</servlet-name>
252         <url-pattern>/login1</url-pattern>
253     </servlet-mapping>
254
255     <servlet>
256         <servlet-name>sessionRetrieve</servlet-name>
257         <servlet-class>SessionRetrieveServlet</servlet-class>
258     </servlet>
259
260     <servlet-mapping>
261         <servlet-name>sessionRetrieve</servlet-name>
262         <url-pattern>/retrieve1</url-pattern>
263     </servlet-mapping>
264
265     <servlet>
266         <servlet-name>logout</servlet-name>
267         <servlet-class>LogoutServlet</servlet-class>
268     </servlet>
269
270     <servlet-mapping>
271         <servlet-name>logout</servlet-name>
272         <url-pattern>/LogoutServlet</url-pattern>
273     </servlet-mapping>
274 </web-app>
275 #SessionLoginServlet.java
276 import java.io.*;
277 import javax.servlet.*;
278 import javax.servlet.http.*;
279
280 public class SessionLoginServlet extends HttpServlet {
281
282     protected void doPost(HttpServletRequest req, HttpServletResponse res)
283         throws IOException {
284
285         res.setContentType("text/html;charset=UTF-8");
286
287         PrintWriter out = res.getWriter();
288
289         String username = req.getParameter("username");
290         String password = req.getParameter("password");
291
292         if (username == null || username.isEmpty()) {
293             out.println("<h3 style='color:red;'>Invalid Login!</h3>");
294             return;
295         }
296
297         HttpSession session = req.getSession();
298
299         session.setAttribute("uname", username);
300         session.setAttribute("pwd", password); // 🔥 ADDED
301
302         session.setMaxInactiveInterval(30 * 60);
303
304         out.println("<h2>Login Successful (Session Created)</h2>");
305         out.println("<a href='retrieve1'>Retrieve Session Details</a>");
306     }
307 }
308
```

```

309 #SessionRetrieveServlet.java
310 import java.io.*;
311 import javax.servlet.*;
312 import javax.servlet.http.*;
313
314 public class SessionRetrieveServlet extends HttpServlet {
315
316     protected void doGet(HttpServletRequest req, HttpServletResponse res)
317         throws IOException {
318
319         res.setContentType("text/html;charset=UTF-8");
320
321         PrintWriter out = res.getWriter();
322
323         HttpSession session = req.getSession(false);
324
325         if (session != null) {
326
327             String user = (String) session.getAttribute("uname");
328             String pass = (String) session.getAttribute("pwd"); // 🔥 ADD
329
330             if (user != null && pass != null) {
331                 out.println("<h2>Welcome " + user + "</h2>");
332                 out.println("<h3>Password: " + pass + "</h3>");
333                 out.println("<a href='LogoutServlet'>Logout</a>");
334             } else {
335                 out.println("<h3>Session exists but data missing</h3>");
336             }
337
338         } else {
339             out.println("<h3>No session found. Please login first.</h3>");
340         }
341     }
342 }
343
344 #LogoutServlet.java
345 import java.io.*;
346 import javax.servlet.*;
347 import javax.servlet.http.*;
348
349 public class LogoutServlet extends HttpServlet {
350
351     protected void doGet(HttpServletRequest req, HttpServletResponse res)
352         throws IOException {
353
354         res.setContentType("text/html;charset=UTF-8");
355
356         PrintWriter out = res.getWriter();
357
358         HttpSession session = req.getSession(false);
359
360         if (session != null) {
361             session.invalidate(); // 🔥 destroy session!!!!!!
362         }
363
364         out.println("<h2>Logged out successfully</h2>");
365         out.println("<a href='session.html'>Login Again</a>");
366     }
367 }

```